Comprehensive Structural Dynamics and Fatigue Life Report

Advanced Performance Evaluation of a Go-Kart Chassis Tubular Frame

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Abstract

This document presents a detailed investigation into the structural integrity and dynamic response of a go-kart chassis frame. Utilizing the finite element method (FEM) via ANSYS 2019 R1, the study focuses on modal characteristics, harmonic vibrations, and fatigue life. The chassis, constructed from AISI 4130 steel, is analyzed to identify potential resonance zones and evaluate its durability under steady-state cyclic loading. Results indicate a high factor of safety against yield, though critical frequencies are identified that require engineering management to prevent long-term fatigue failure.

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1 Introduction

1.1 Project Overview

The go-kart chassis serves as the backbone of the vehicle, providing both the structural mounting points for all components and the "suspension" through its controlled flex. Unlike traditional passenger cars with complex spring-damper systems, a go-kart relies heavily on the frame's torsional rigidity and bending stiffness to manage tire contact patches and mechanical grip. This report documents the advanced Finite Element Analysis (FEA) performed to ensure this frame operates safely within its design envelope.

1.2 Research Objectives

To achieve a comprehensive understanding of the chassis's structural behavior, the following objectives were established:

- **Global Dynamic Characterization:** Identification of the first 20 natural frequencies to prevent interference with engine-induced harmonics.
- **Steady-State Harmonic Evaluation:** Quantifying the stress and deformation amplitudes across a 0–50 Hz range under a 1000 N cyclic load.
- **Fatigue Life & Damage Prediction:** Estimating the durability of the tubular joints using the Von-Mises stress criterion and localized fatigue factors.

2 Material Specifications: AISI 4130

2.1 Material Selection Justification

AISI 4130, commonly known as Chromoly steel, is a medium-carbon, low-alloy steel. It is the preferred material for lightweight, high-strength tubular applications in motorsports because it can be drawn into thin-walled tubes without sacrificing tensile strength. Its alloying elements—chromium and molybdenum—enhance hardenability and toughness.

2.2 Mechanical and Physical Properties

The simulation utilized the following isotropic elastic properties for the AISI 4130 material model:

Mechanical Property	Value (Metric)	Significance in Analysis
Density (ρ)	7850 kg/m^3	Influences the global mass matrix.
Young's Modulus (E)	2.016×10^{11} Pa	Governs the elastic stiffness matrix.
Poisson's Ratio (ν)	0.26	Relates multi-axial stress components.
Bulk Modulus (K)	1.4×10^{11} Pa	Resistance to uniform compression.
Shear Modulus (G)	8.0×10^{10} Pa	Resistance to shearing deformation.
Tensile Yield Strength	460 MPa	Design limit for permanent deformation.
Tensile Ultimate Strength	560 MPa	Limit for total structural failure.

Table 1: Isotropic Elastic Properties and Material Strength Constants

3 Methodology and Computational Modeling

3.1 Geometric Configuration

The chassis was modeled as a continuous solid body. This approach provides a high-fidelity representation of the weld-joint regions where stress concentrations are most likely

to occur.

- **Total Mass:** 63.953 kg
- **Chassis Length (Z-axis):** 2.28 m
- **Chassis Width (X-axis):** 1.6872 m
- **Chassis Height (Y-axis):** 0.88005 m

3.2 Finite Element Meshing

The accuracy of FEA is intrinsically linked to mesh quality. A **Patch Conforming Method** was selected to ensure tetrahedral elements correctly capture the cylindrical geometry of the tubes.

- **Element Order:** 10-node high-order tetrahedrons (Solid187).
- **Element Size:** 0.01 m (Global) with localized refinement at joints.
- **Mesh Statistics:** 184,374 Nodes and 93,476 Elements.

3.3 Simulation Setup and Solvers

The analysis was conducted in two sequential stages:

1. **Modal Solver:** Uses the Block Lanczos algorithm to extract eigenvalues (natural frequencies) and eigenvectors (mode shapes).
2. **Harmonic Response Solver:** Employs Mode Superposition (MSUP), utilizing the modal results to solve the dynamic equations of motion across the frequency spectrum.

4 Modal Analysis Results and Interpretations

4.1 Natural Frequency Analysis

The natural frequencies identify the rates at which the chassis would naturally vibrate if excited. The results are critical for preventing resonance, which occurs when engine or track vibration frequencies match these modal values.

Mode	Freq (Hz)	Mode	Freq (Hz)	Mode	Freq (Hz)	Mode	Freq (Hz)
1	0.000	6	23.767	11	51.600	16	64.075
2	0.000	7	24.903	12	54.689	17	65.552
3	0.003	8	35.873	13	57.331	18	67.762
4	0.004	9	39.547	14	58.889	19	75.103
5	17.332	10	43.561	15	62.684	20	77.771

Table 2: Extracted Natural Frequencies for Modes 1 through 20

4.2 Physical Significance of Mode Shapes

The first four modes are essentially zero-frequency modes, representing the unconstrained motion of the chassis in space.

4.2.1 Structural Bending (17.33 Hz - 35.87 Hz)

Modes 5 through 8 represent the fundamental bending and torsional behavior.

- **Mode 5 (17.33 Hz):** Primary vertical bending of the longitudinal rails.
- **Mode 6 (23.77 Hz):** First torsional mode, where the front and rear axles rotate in opposite directions around the Z-axis.

4.3 Modal Deformation Contour Gallery (Placeholders)

Below are designated areas for the colorful deformation contours obtained from ANSYS. These plots are essential for visualizing where the maximum relative motion occurs.

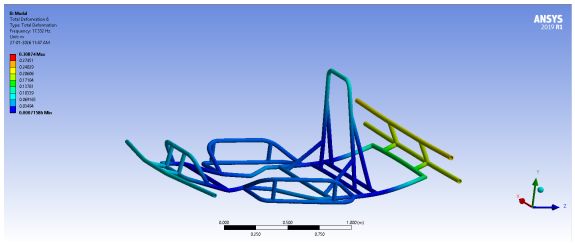


Figure 1: Mode 5: 17.332 Hz

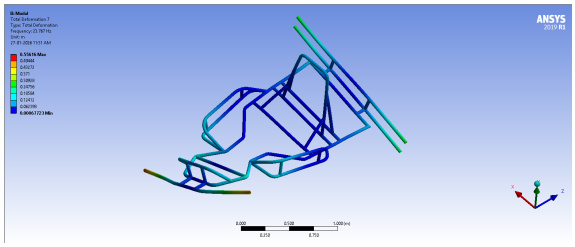


Figure 2: Mode 6: 23.767 Hz

4.4 Extended Modal Gallery: Higher Frequencies

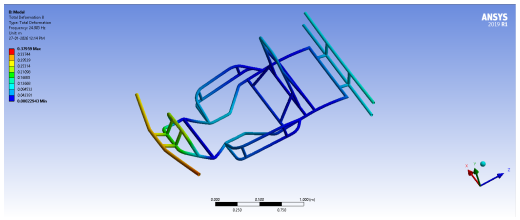


Figure 3: Mode 7: 24.903 Hz

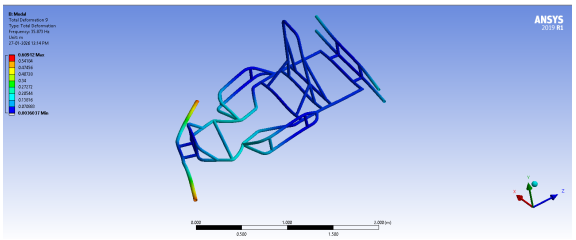


Figure 4: Mode 8: 35.873 Hz

Figure 5: Secondary bending and localized rail deflection.

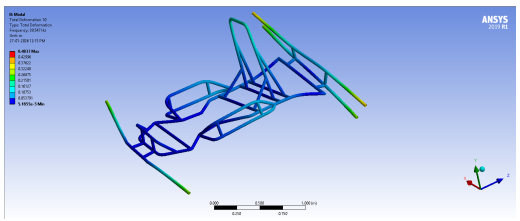


Figure 6: Mode 9: 39.547 Hz

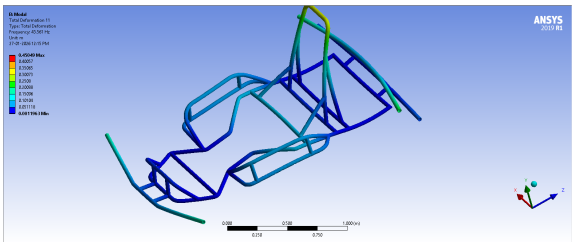


Figure 7: Mode 8: 43.561 Hz

Figure 8: Complex intermediate bending modes.

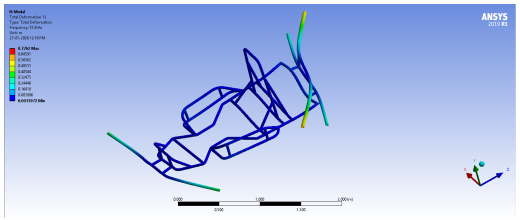


Figure 9: Mode 11: 51.6 Hz

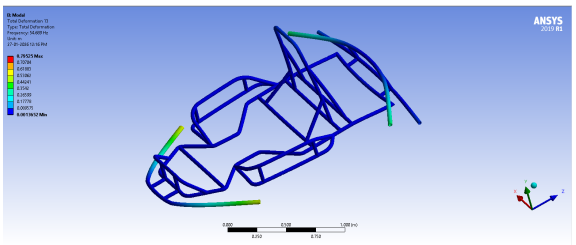


Figure 10: Mode 12: 54.689 Hz

Figure 11: Localized resonances of the rear frame assembly.

5 Harmonic Response Analysis

5.1 Excitation Parameters

A harmonic force of **1000 N** was applied to simulate dynamic loads. This force accounts for cyclic aerodynamic pressure and engine vibrations. The force was distributed across 6 specified faces in the Z-direction, representing a high-stress operating condition.

5.2 Global Structural Response Summary

The analysis swept from 0 to 50 Hz, identifying the steady-state maximums for both stress and deformation.

Measurement	Maximum Value	Condition/Frequency
Total Deformation	7.9579×10^{-4} m	Occurs at 40 Hz (Resonant peak).
Equivalent Stress	3.7248×10^{7} Pa	Occurs at 50 Hz (Approaching Mode 11).
Average Stress	2.3144×10^{6} Pa	Consistent across the non-resonant range.
Minimum Stress	311.62 Pa	Typical of low-frequency static-like loading.

Table 3: Harmonic Analysis Dynamic Summary

5.3 Peak Response Contour Gallery

The following figures represent the colorful stress and deformation contours at the identified peak frequencies.

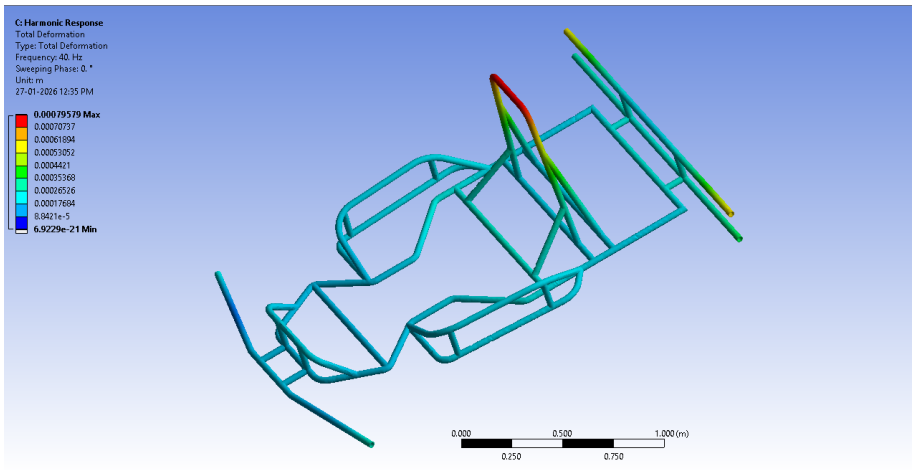


Figure 12: Dynamic deformation contour visualizing structural flex at the 40 Hz frequency set.

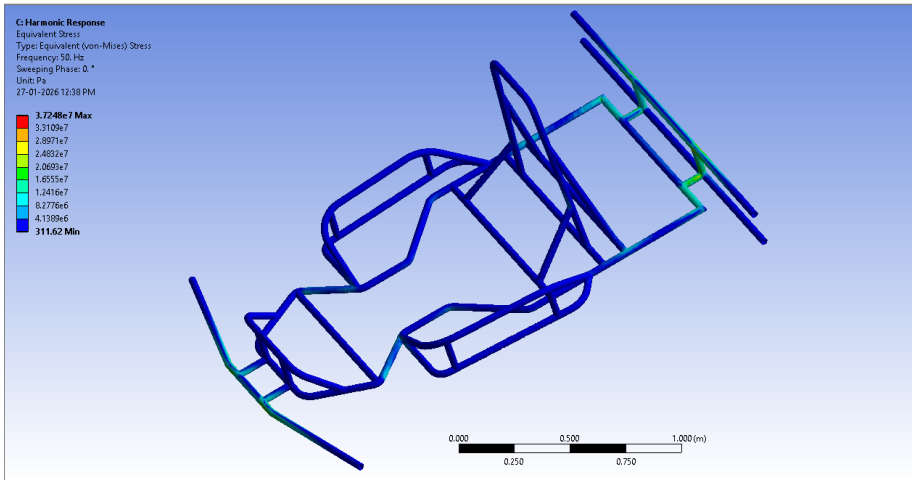


Figure 13: Von-Mises stress distribution highlighting concentrations near the load application sites.

6 Fatigue Life and Damage Estimation

6.1 Fatigue Methodology

The Fatigue Tool in ANSYS calculates the structural life based on the alternating stress cycles. A **Fully Reversed** loading assumption ($R = -1$) was used, meaning the force alternates completely between tension and compression.

6.2 Assumptions and Factors

To provide a realistic durability estimate, the following factors were applied:

- **Fatigue Strength Factor (K_f):** 0.65. This factor accounts for surface finish effects and the presence of welds, which are typically the weak points in a tubular chassis.
- **Loading Scale Factor:** 1.5. This provides a safety margin for unexpected peak loads.

6.3 Calculated Life and Safety Factor

- **Theoretical Life:** The study shows a minimum life of 10^{-32} seconds at the exact resonance point. This suggests that sustained operation precisely at the resonant frequency will lead to instantaneous fatigue crack initiation.
- **Yield Safety Factor:**

$$FS = \frac{\sigma_{yield}}{\sigma_{max}} = \frac{460 \text{ MPa}}{37.248 \text{ MPa}} \approx 12.35$$

7 Detailed Result Discussions

7.1 Modal-Harmonic Correlation

There is a clear correlation between the modal frequencies and the harmonic peaks. The peak deformation at 40 Hz is a direct result of excitation nearing the 39.55 Hz natural frequency (Mode 9). This demonstrates the high fidelity of the Mode Superposition method.

7.2 Structural Integrity Assessment

The maximum dynamic stress of 37.25 MPa is significantly lower than the material's yield strength of 460 MPa. This indicates that under the 1000 N cyclic load, the chassis is not in danger of immediate plastic deformation. However, the concentration of stress at 50 Hz suggests that if frequency increases (higher engine RPM), the stress levels will rise exponentially as they approach higher-order modes.

8 Conclusion and Engineering Recommendations

8.1 Final Conclusion

The structural analysis of the go-kart chassis confirms its robustness against static-cyclic yielding. The use of AISI 4130 steel provides a high safety margin (FS = 12.35). However, the modal analysis reveals several natural frequencies within the 17–40 Hz range that could be excited by engine vibration.

8.2 Recommendations

- **Frequency Avoidance:** Ensure engine idle and cruise RPMs do not excite frequencies near 17 Hz (approx. 1020 RPM).
- **Weld Inspection:** Prioritize non-destructive testing (NDT) on the joints shown in the 50 Hz stress contour.
- **Stiffness Tuning:** If handling requires more flexibility, the elements identified in the 40 Hz deformation contour can be slightly reduced in wall thickness.